

Weapon Detection Using YOLO

● **About the model**

The aim of this weapon detection project is to develop an effective deep learning-based solution for identifying weapons in the hands of individuals. The project utilizes the Ultralytics YOLO (You Only Look Once) model, a state-of-the-art real-time object detection algorithm, to accurately detect and localize weapons in images or video frames. YOLO's speed and accuracy make it an ideal choice for real-time applications like this one.

The project involves setting up the environment, training the YOLO model on a custom dataset, evaluating its performance, and making predictions on test images. The ultimate goal is to enhance security and safety by automating the detection of potentially dangerous objects in real-world scenarios.

● **Project Details**

→ **Checking GPU Availability**

This command checks whether a compatible NVIDIA GPU is available for accelerated deep learning model training. GPUs are essential for efficiently training deep neural networks.

```
!nvidia-smi
```

Tue Sep 19 11:38:36 2023

NVIDIA-SMI 525.105.17 Driver Version: 525.105.17 CUDA Version: 12.0									
GPU	Name	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC		
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage	GPU-Util	Compute M.	MIG M.		
0	Tesla T4	Off	00000000:00:04.0	Off			0		
N/A	65C	P8	11W / 70W	0MiB / 15360MiB	0%	Default	N/A		

Processes:									
GPU	GI	CI	PID	Type	Process name	GPU Memory			
	ID	ID				Usage			
No running processes found									

→ Setting Up the Environment

Here we get the current working directory and store it in the HOME variable. We then print the current working directory.

```
import os
HOME=os.getcwd()
print(HOME)
```

/content/drive/MyDrive/FINAL_GS_weapon2

→ Installing Ultralytics

Here, we install the Ultralytics library, which is a popular framework for working with YOLO computer vision models. It provides tools and utilities for training and deploying object detection models.

```
!pip install ultralytics

Collecting ultralytics
  Downloading ultralytics-8.0.227-py3-none-any.whl (660 kB)
    660.5/660.5 kB 4.0 MB/s eta 0:00:00
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (3.7.1)
Requirement already satisfied: numpy>=1.22.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.23.5)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (4.8.0.76)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (9.4.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (6.0.1)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.31.0)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.11.4)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (2.1.0+cu121)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (0.16.0+cu121)
Requirement already satisfied: tqdm>=4.64.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (4.66.1)
Requirement already satisfied: pandas>=1.1.4 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (1.5.3)
Requirement already satisfied: seaborn>=0.11.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics) (0.12.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.10/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.10/dist-packages (from ultralytics) (9.0.0)
Collecting thop>=0.1.1 (from ultralytics)
```

→ Importing and Checking Ultralytics

In this part, we import the Ultralytics library and use the `clear_output` function to clear the current output. Then, we run checks to ensure that Ultralytics is correctly installed and configured.

```
[3]
from IPython import display
display.clear_output()

import ultralytics
ultralytics.checks()

Ultralytics YOLOv8.0.227 🚀 Python-3.10.12 torch-2.1.0+cu121 CPU (Intel Xeon 2.20GHz)
Setup complete ✅ (2 CPUs, 12.7 GB RAM, 26.2/107.7 GB disk)
```

→ Importing YOLO from Ultralytics & Mounting Google Drive

We import the YOLO class from Ultralytics. This class is used for creating and managing YOLO models for object detection tasks.

In the next part we mount Google Drive into the Colab environment. It's a common practice for accessing data and project files stored in your Google Drive. We then navigate to the directory where your project files are located.

```
✓ [4] from ultralytics import YOLO
0s      from IPython.display import display, Image

[ ] Start coding or generate with AI.
```

```
✓ %cd /content/drive/MyDrive/FINAL_GS_weapon2
0s  [icon] /content/drive/MyDrive/FINAL_GS_weapon2
```

```
✓ [7] from google.colab import drive
0s      drive.mount('/content/drive')

Drive already mounted at /content/drive; to
```

→ Model Training

In this step, we start training a YOLO model. The command specifies various parameters, such as the training mode, the YOLO model architecture (yolov8s), the data configuration file (data.yaml), the number of training epochs (25), the input image size (imgsz), and whether to generate training plots (plots=True).

```
!yolo task=detect mode=train model=yolov8s.pt data= data.yaml epochs=25 imgsz=224 plots=True
```

```
ultralytics YOLOv8.0.227 Python-3.10.12 torch-2.1.0+cu121 CPU (Intel Xeon 2.20GHz)
engine/trainer: task=detect, mode=train, model=yolov8s.pt, data=data.yaml, epochs=25, patience=50, batch=16, imgsz=224, save=True, save_period=-1, cache=False, device=None, workers=
2023-12-15 14:03:14.648661: E external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:9261] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when on
2023-12-15 14:03:14.648740: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:607] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one
2023-12-15 14:03:14.653416: E external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1515] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when
Overriding model.yaml nc=80 with nc=1
```

	from	n	params	module	arguments
0	-1	1	928	ultralytics.nn.modules.conv.Conv	[3, 32, 3, 2]
1	-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
2	-1	1	29056	ultralytics.nn.modules.block.C2f	[64, 64, 1, True]
3	-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
4	-1	2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
5	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
6	-1	2	788480	ultralytics.nn.modules.block.C2f	[256, 256, 2, True]
7	-1	1	1180672	ultralytics.nn.modules.conv.Conv	[256, 512, 3, 2]
8	-1	1	1838080	ultralytics.nn.modules.block.C2f	[512, 512, 1, True]
9	-1	1	656896	ultralytics.nn.modules.block.SPPF	[512, 512, 5]
10	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
12	-1	1	591360	ultralytics.nn.modules.block.C2f	[768, 256, 1]
13	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
14	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
15	-1	1	148224	ultralytics.nn.modules.block.C2f	[384, 128, 1]
16	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
17	[-1, 12]	1	0	ultralytics.nn.modules.conv.Concat	[1]
18	-1	1	493056	ultralytics.nn.modules.block.C2f	[384, 256, 1]
19	-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]
20	[-1, 9]	1	0	ultralytics.nn.modules.conv.Concat	[1]
21	-1	1	1969152	ultralytics.nn.modules.block.C2f	[768, 512, 1]

```
✓ 0s completed at 21:41
```

Transferred 349/355 items from pretrained weights

TensorBoard: Start with 'tensorboard --logdir runs/detect/train2', view at <http://localhost:6006/>

Freezing layer 'model.22.dfl.conv.weight'

train: Scanning /content/drive/MyDrive/FINAL_GS_weapon2/weapon_data/labels/train.cache... 437 images, 9 backgrounds, 0 corrupt

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01), CLAHE(p=0.01, clip_

val: Scanning /content/drive/MyDrive/FINAL_GS_weapon2/weapon_data/labels/val.cache... 30 images, 0 backgrounds, 0 corrupt:

Plotting labels to runs/detect/train2/labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr' and 'mo

optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay

Image sizes 224 train, 224 val

Using 0 dataloader workers

Logging results to runs/detect/train2

Starting training for 25 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/25	0G	1.904	2.282	1.581	41	224: 100% 28/28 [02:23<00:00, 5.13s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:05<00:00, 5.30s/it]
	all	30	47	0.3	0.426	0.3 0.116

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/25	0G	1.891	1.895	1.509	40	224: 100% 28/28 [02:24<00:00, 5.18s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:03<00:00, 3.82s/it]
	all	30	47	0.232	0.298	0.211 0.0605

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
3/25	0G	1.91	1.944	1.585	62	224: 100% 28/28 [02:20<00:00, 5.03s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:03<00:00, 3.79s/it]
	all	30	47	0.272	0.191	0.136 0.0403

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
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0s completed at 21:41

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
3/25	0G	1.91	1.944	1.585	62	224: 100% 28/28 [02:20<00:00, 5.03s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:03<00:00, 3.79s/it]
	all	30	47	0.272	0.191	0.136 0.0403

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
4/25	0G	1.954	1.967	1.637	43	224: 100% 28/28 [02:17<00:00, 4.93s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:03<00:00, 3.74s/it]
	all	30	47	0.00282	0.191	0.00115 0.000427

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
5/25	0G	1.959	1.906	1.608	41	224: 100% 28/28 [02:21<00:00, 5.05s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:03<00:00, 3.77s/it]
	all	30	47	0.00355	0.0851	0.00118 0.000553

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
6/25	0G	1.936	1.869	1.621	31	224: 100% 28/28 [02:16<00:00, 4.88s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:04<00:00, 4.48s/it]
	all	30	47	0.0306	0.0638	0.0131 0.00512

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
7/25	0G	1.95	1.882	1.626	35	224: 100% 28/28 [02:18<00:00, 4.93s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:03<00:00, 3.52s/it]
	all	30	47	0.0894	0.128	0.0618 0.024

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
8/25	0G	1.884	1.805	1.546	32	224: 100% 28/28 [02:16<00:00, 4.89s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 1/1 [00:04<00:00, 4.52s/it]
	all	30	47	0.46	0.255	0.217 0.0845

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
9/25	0G	1.839	1.768	1.542	35	224:	100% 28/28 [02:18<00:00, 4.96s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.49s/it]
	all	30	47	0.521	0.298	0.364	0.147
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
10/25	0G	1.778	1.666	1.518	37	224:	100% 28/28 [02:16<00:00, 4.87s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:04<00:00, 4.92s/it]
	all	30	47	0.571	0.383	0.409	0.149
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
11/25	0G	1.789	1.652	1.494	25	224:	100% 28/28 [02:19<00:00, 4.97s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:04<00:00, 4.94s/it]
	all	30	47	0.8	0.383	0.467	0.192
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
12/25	0G	1.744	1.631	1.498	37	224:	100% 28/28 [02:17<00:00, 4.90s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.55s/it]
	all	30	47	0.644	0.385	0.471	0.152
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
13/25	0G	1.719	1.553	1.458	28	224:	100% 28/28 [02:21<00:00, 5.06s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:04<00:00, 4.09s/it]
	all	30	47	0.533	0.383	0.437	0.17
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
14/25	0G	1.62	1.438	1.397	43	224:	100% 28/28 [02:15<00:00, 4.85s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.67s/it]
	all	30	47	0.605	0.447	0.469	0.2
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
15/25	0G	1.604	1.451	1.424	37	224:	100% 28/28 [02:17<00:00, 4.89s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.57s/it]
	all	30	47	0.718	0.404	0.503	0.21

✓ 0s completed at 21:41

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
16/25	0G	1.551	1.411	1.417	18	224:	100% 28/28 [02:18<00:00, 4.95s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:04<00:00, 4.40s/it]
	all	30	47	0.556	0.574	0.543	0.261
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
17/25	0G	1.457	1.24	1.376	20	224:	100% 28/28 [02:20<00:00, 5.02s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.68s/it]
	all	30	47	0.552	0.617	0.558	0.271
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
18/25	0G	1.433	1.201	1.358	21	224:	100% 28/28 [02:17<00:00, 4.89s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.64s/it]
	all	30	47	0.758	0.489	0.568	0.253
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
19/25	0G	1.384	1.097	1.308	22	224:	100% 28/28 [02:16<00:00, 4.89s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:04<00:00, 4.93s/it]
	all	30	47	0.869	0.447	0.533	0.229
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
20/25	0G	1.332	1.016	1.294	20	224:	100% 28/28 [02:15<00:00, 4.86s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.54s/it]
	all	30	47	0.844	0.383	0.502	0.254
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
21/25	0G	1.334	0.9976	1.281	17	224:	100% 28/28 [02:17<00:00, 4.90s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:04<00:00, 4.30s/it]
	all	30	47	0.698	0.442	0.509	0.253
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
22/25	0G	1.26	0.9659	1.232	38	224:	100% 28/28 [02:20<00:00, 5.02s/it]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 1/1 [00:03<00:00, 3.50s/it]

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:04<00:00, 4.52s/it]
	all	30	47	0.869	0.447	0.533	0.229	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size		
20/25	0G	1.332	1.016	1.294	20	224:	100% 28/28 [02:15<00:00, 4.86s/it]	
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:03<00:00, 3.54s/it]
	all	30	47	0.844	0.383	0.502	0.254	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size		
21/25	0G	1.334	0.9976	1.281	17	224:	100% 28/28 [02:17<00:00, 4.90s/it]	
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:04<00:00, 4.30s/it]
	all	30	47	0.698	0.442	0.509	0.253	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size		
22/25	0G	1.26	0.9659	1.232	38	224:	100% 28/28 [02:20<00:00, 5.02s/it]	
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:03<00:00, 3.50s/it]
	all	30	47	0.723	0.445	0.527	0.268	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size		
23/25	0G	1.22	0.893	1.221	16	224:	100% 28/28 [02:17<00:00, 4.92s/it]	
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:04<00:00, 4.30s/it]
	all	30	47	0.664	0.504	0.544	0.255	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size		
24/25	0G	1.203	0.8589	1.196	29	224:	100% 28/28 [02:18<00:00, 4.94s/it]	
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:04<00:00, 4.82s/it]
	all	30	47	0.752	0.451	0.487	0.262	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size		
25/25	0G	1.133	0.7828	1.148	18	224:	100% 28/28 [02:16<00:00, 4.87s/it]	
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:03<00:00, 3.71s/it]
	all	30	47	0.742	0.49	0.576	0.284	

25 epochs completed in 1.001 hours.

Optimizer stripped from runs/detect/train2/weights/last.pt, 22.5MB

Optimizer stripped from runs/detect/train2/weights/best.pt, 22.5MB

Validating runs/detect/train2/weights/best.pt...

Ultralytics YOLOv8.0.227 🚀 Python-3.10.12 torch-2.1.0+cu121 CPU (Intel Xeon 2.20GHz)

Model summary (fused): 168 layers, 11125971 parameters, 0 gradients, 28.4 GFLOPs

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)	100% 1/1 [00:04<00:00, 4.52s/it]
all	30	47	0.742	0.49	0.576	0.284	

Speed: 0.4ms preprocess, 133.1ms inference, 0.0ms loss, 0.3ms postprocess per image

Results saved to **runs/detect/train2**

💡 Learn more at <https://docs.ultralytics.com/modes/train>

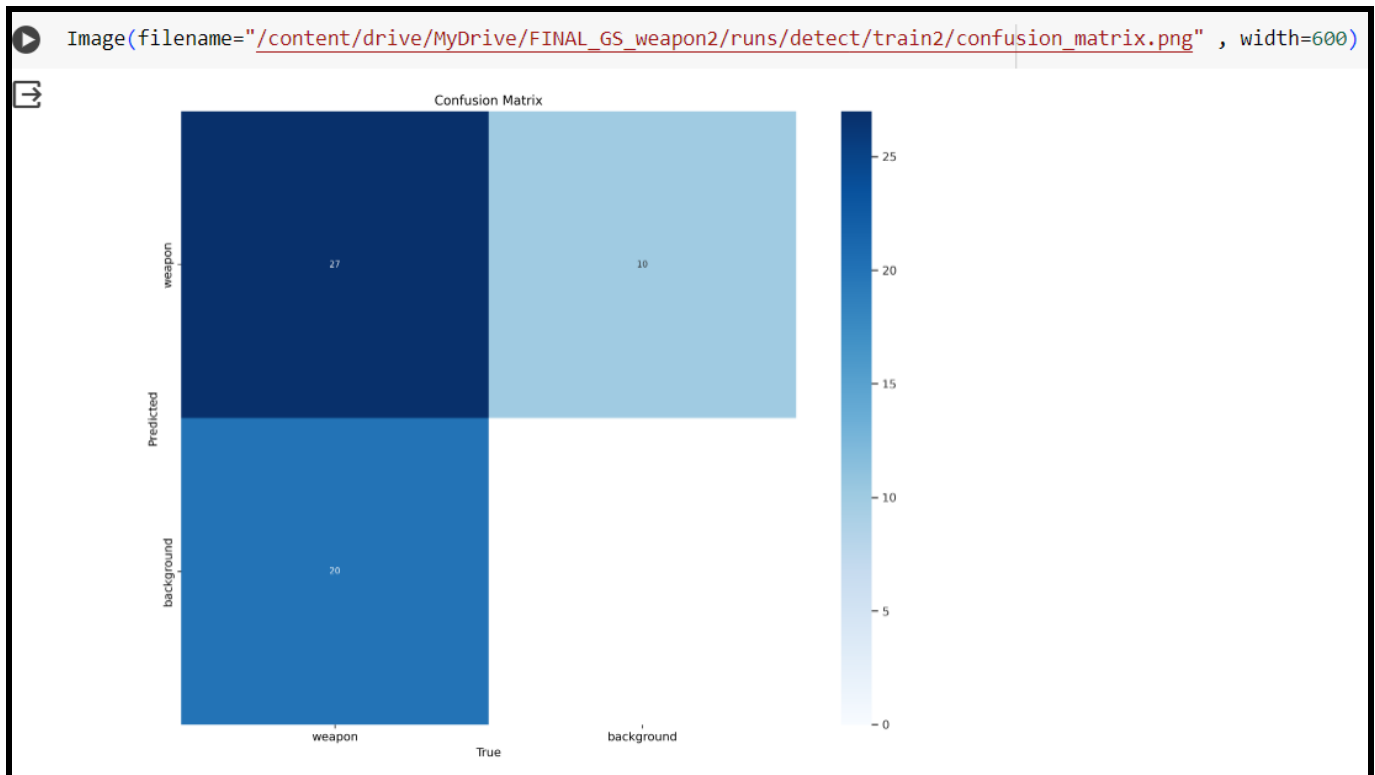
→ Listing Training Results

After training the model, we list the contents of the "runs/detect/train6/" directory. This contains various training-related files and artifacts, such as model weights, logs, and metrics.

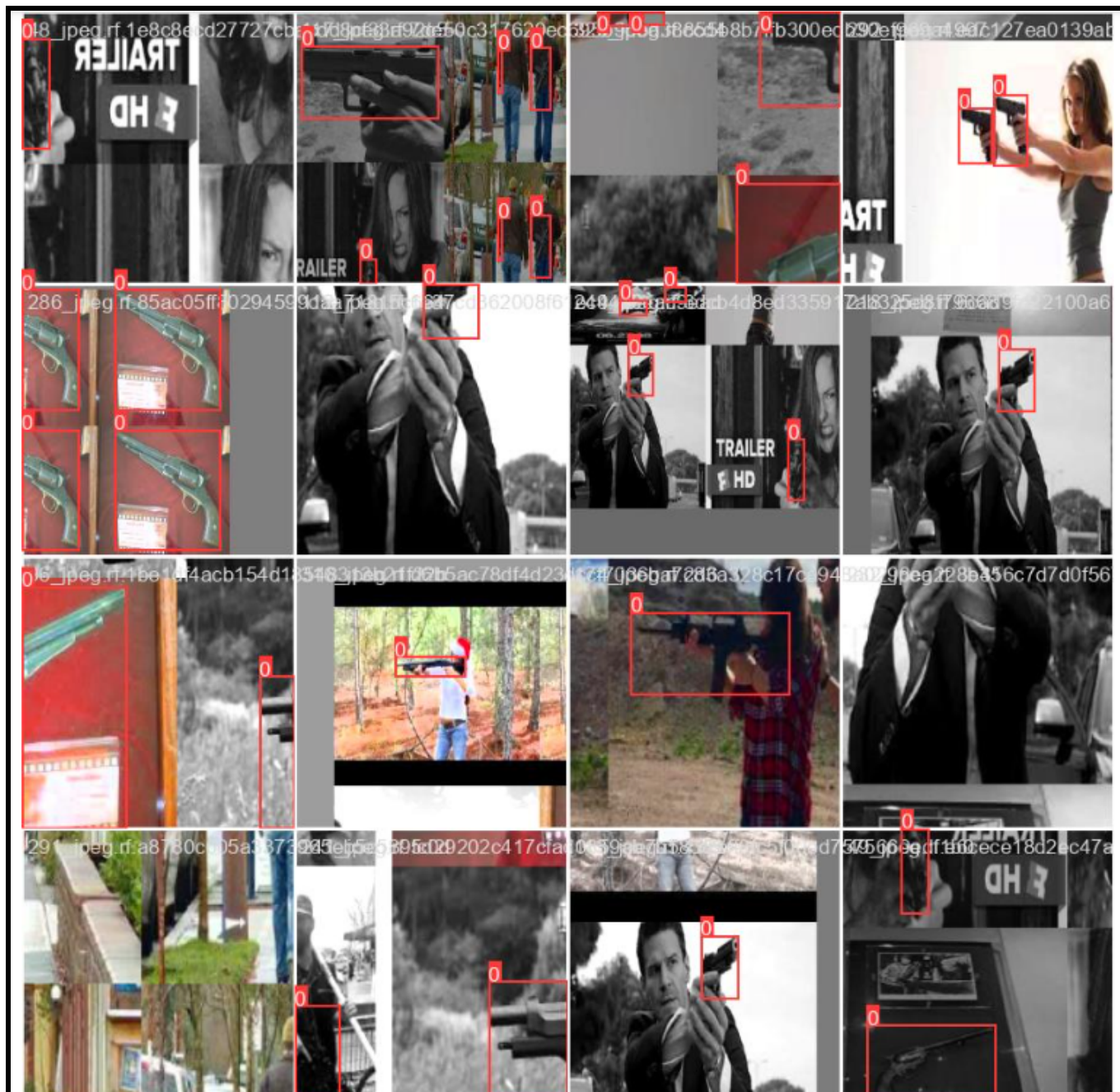
```
✓ [17] !ls runs/detect/train2/
0s
args.yaml
confusion_matrix_normalized.png
confusion_matrix.png
events.out.tfevents.1702648999.75cd95999981.6856.0
F1_curve.png
labels_correlogram.jpg
labels.jpg
P_curve.png
PR_curve.png
R_curve.png
results.csv
results.png
train_batch0.jpg
train_batch1.jpg
train_batch2.jpg
train_batch420.jpg
train_batch421.jpg
train_batch422.jpg
val_batch0_labels.jpg
val_batch0_pred.jpg
weights
```

→ Visualizing Training Metrics

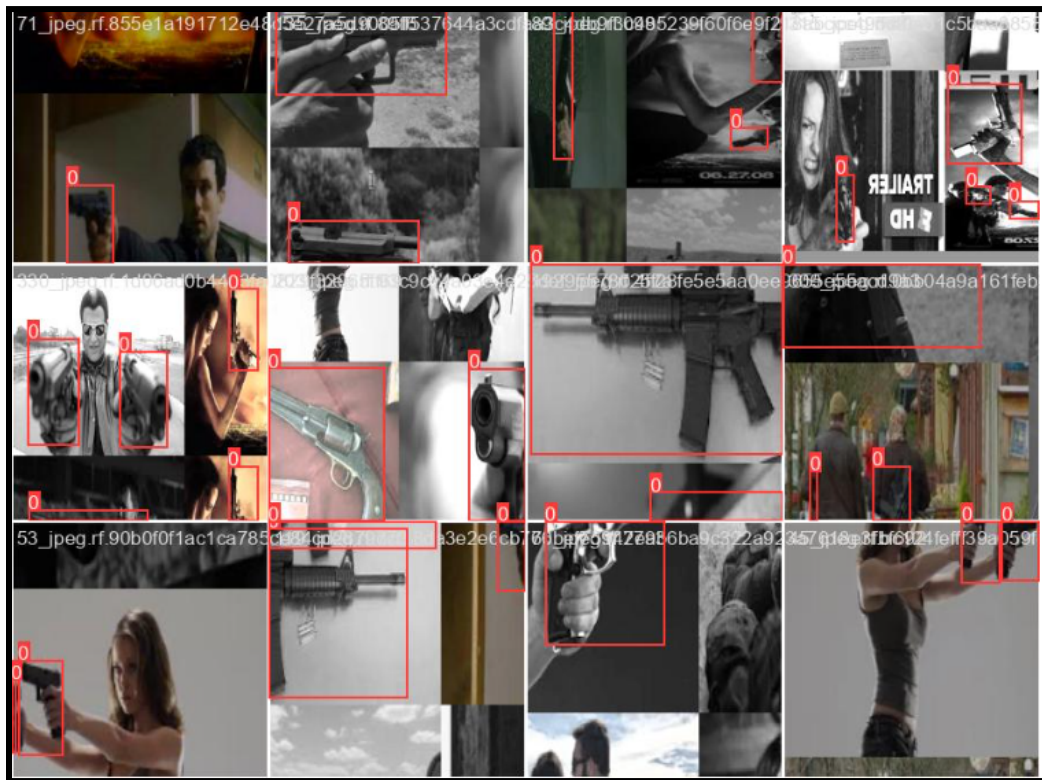
Here, we display visualizations of the training metrics. The first image is a confusion matrix, which provides insights into the model's performance. The second image is a training result from the train dataset.



Image(filename='/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train2/train_batch0.jpg', width=600)



Image(filename='/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train3/train_batch1.jpg', width=600)



→ Model Evaluation

We evaluate the trained YOLO model on the validation dataset using the model weights stored in "best.pt." The data configuration file (data.yaml) is used for this evaluation.

```
!yolo task=detect mode=val model=/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train2/weights/best.pt data=data.yaml
```

Ultralytics YOLOv8.0.227 Python-3.10.12 torch-2.1.0+cu121 CPU (Intel Xeon 2.20GHz)
Model summary (fused): 168 layers, 11125971 parameters, 0 gradients, 28.4 GFLOPs
val: Scanning /content/drive/MyDrive/FINAL_GS_weapon2/weapon_data/labels/val.cache... 30 images, 0 backgrounds, 0 corrupt: 100% 30/30

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)
all	30	47	0.742	0.49	0.576	0.284

Speed: 2.5ms preprocess, 223.1ms inference, 0.0ms loss, 0.3ms postprocess per image
Results saved to runs/detect/val
Learn more at <https://docs.ultralytics.com/modes/val>

→ Making Predictions

In this step, we use the trained model to make predictions on a test image. We specify the model weights, a confidence threshold (conf=0.25), and the source image for prediction.

```
!yolo task=detect mode=predict model=/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train2/weights/best.pt conf=0.25 source=/content/drive/MyDrive/FINAL_GS_weapon2/weapon_data/images/test/116_jpeg.rf.182ed1b5b3e12352ac39e03db712d5f5.jpg
```

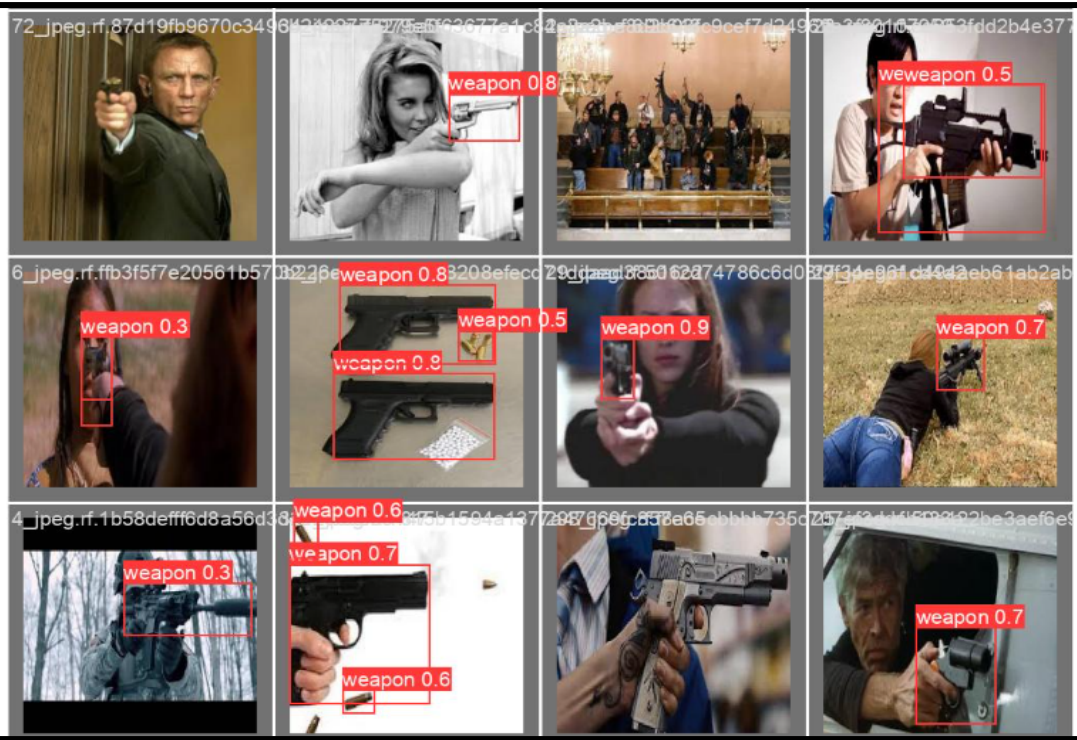
Ultralytics YOLOv8.0.227 Python-3.10.12 torch-2.1.0+cu121 CPU (Intel Xeon 2.20GHz)
Model summary (fused): 168 layers, 11125971 parameters, 0 gradients, 28.4 GFLOPs

image 1/1 /content/drive/MyDrive/FINAL_GS_weapon2/weapon_data/images/test/116_jpeg.rf.182ed1b5b3e12352ac39e03db712d5f5.jpg: 224x224 1 weapon, 104.1ms
Speed: 1.0ms preprocess, 104.1ms inference, 1.4ms postprocess per image at shape (1, 3, 224, 224)
Results saved to runs/detect/predict
Learn more at <https://docs.ultralytics.com/modes/predict>

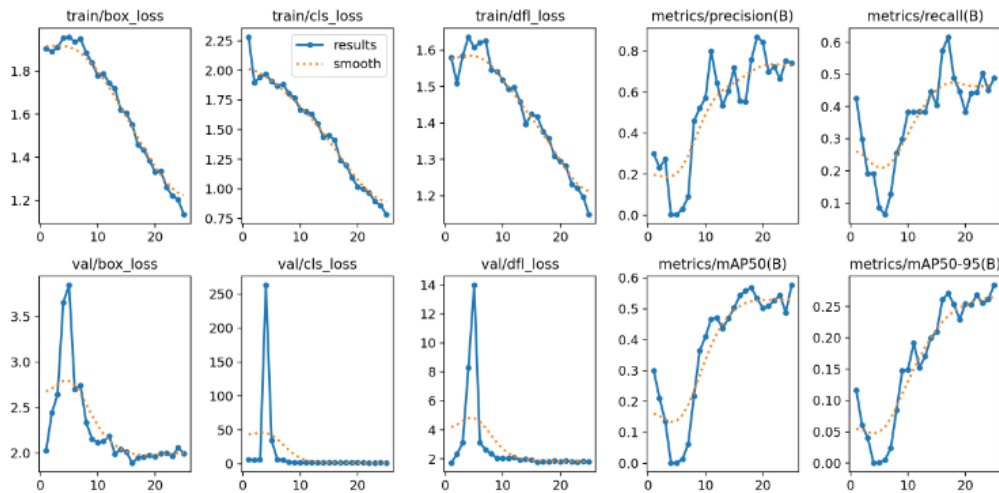
→ Additional Visualization

We display additional visualizations, including prediction results and a summary of the overall model performance.

Image(filename='/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train3/val_batch0_pred.jpg', width=600)



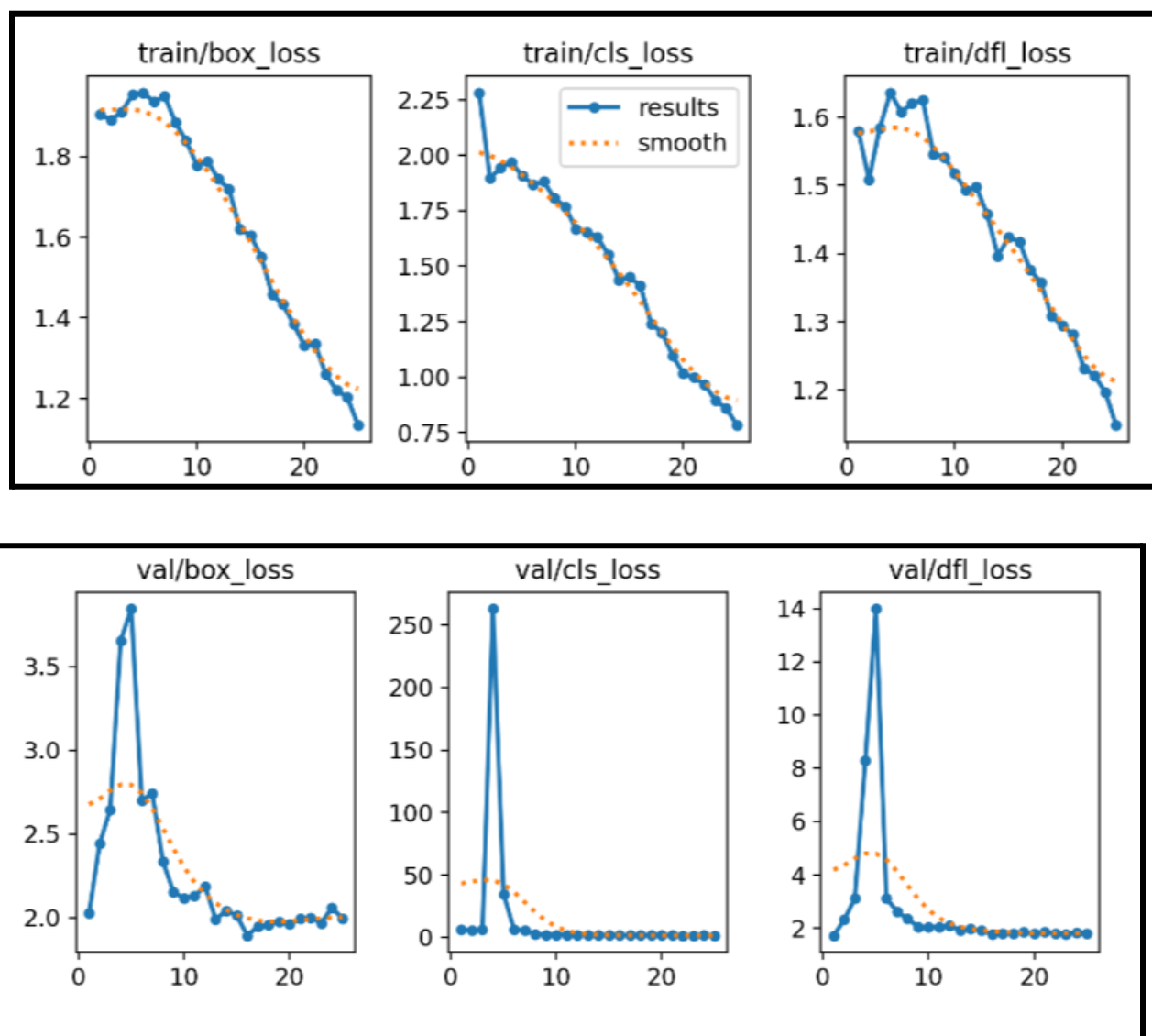
```
Image(filename='/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train2/results.png', width=600)
```



→ Accuracy Metric Curves

- **box_loss:** This curve represents the error in predicting the bounding boxes around detected weapons. The y-axis shows the magnitude of loss, and x-axis indicates epochs or iterations. A lower value indicates that the model is getting better at accurately drawing boxes around weapons.
- **cls_loss:** This curve illustrates classification loss, indicating how well the model is classifying objects within those predicted bounding boxes as weapons or non-weapons. A decrease in this value over epochs (x-axis) means improved accuracy in classification.
- **dfl_loss:** This curve represents the focal loss, which is a modification of the standard cross-entropy loss that is used to address class imbalance in object detection tasks. The y-axis shows the magnitude of loss, and x-axis indicates epochs or iterations. A

lower value indicates that the model is getting better at detecting weapons.



- **Precision Curve (Top Left):** This graph shows how precise the model is, meaning how many of the detected objects are actual weapons. A higher precision indicates fewer false positives.

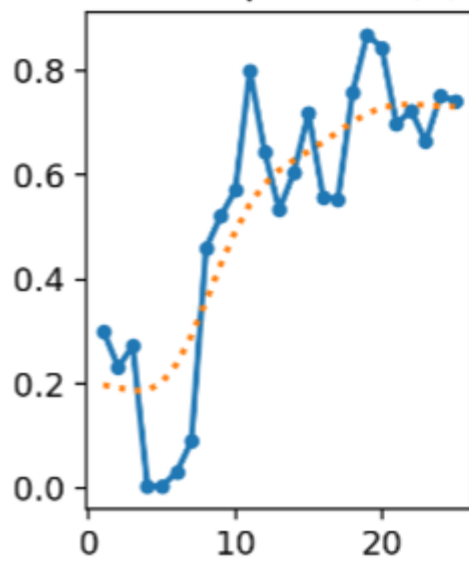
$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

- **Recall Curve (Top Right):** This curve represents the recall rate, indicating how many actual weapons in the images were detected by the model. A higher recall means that the model is good at identifying weapons but may also include some false positives.

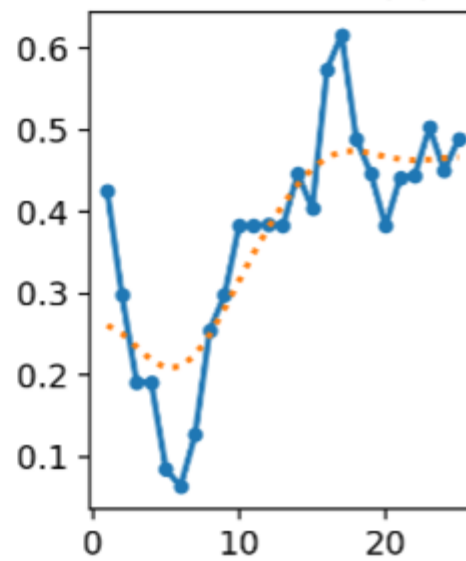
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

- **mAP50 Curve (Bottom Left):** mAP50 stands for mean Average Precision at 50% IoU (Intersection over Union). It gives us an idea about the accuracy of our model; higher values indicate better performance in terms of both precision and recall but only considering detections as true positives if they have an IoU above 50% with ground truths.
- **mAP50-95 Curve (Bottom Right):** This is a metric which takes into account detections as true positives only if they have an IoU between 50% and 95% with ground truths. It provides a broader perspective on the model's performance across different levels of detection strictness.

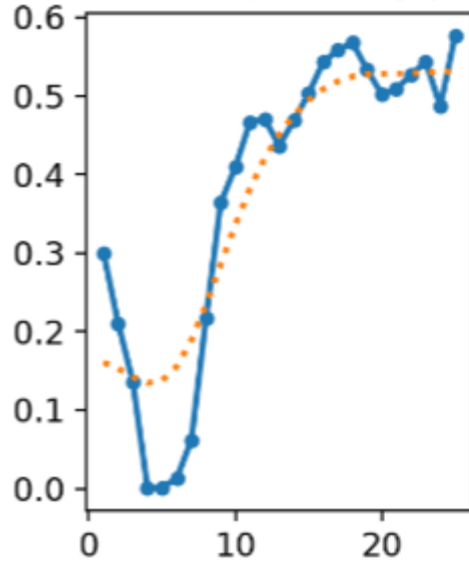
metrics/precision(B)



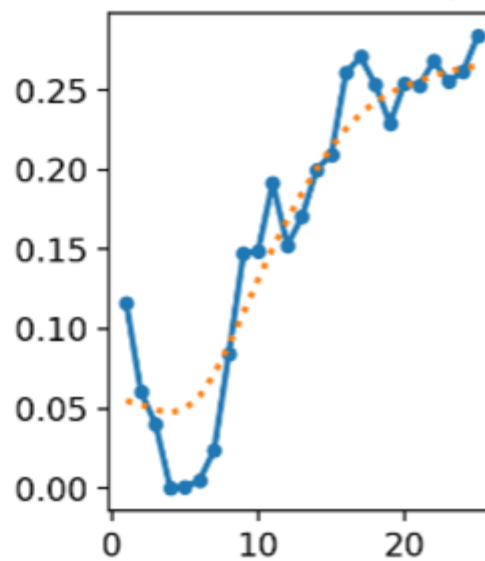
metrics/recall(B)



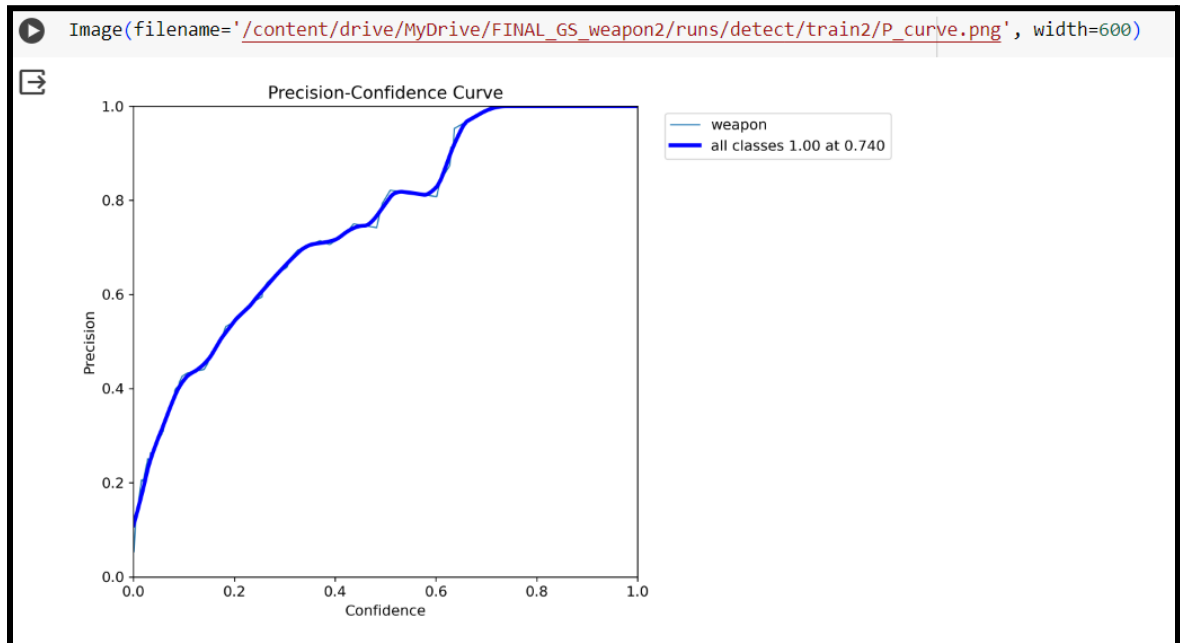
metrics/mAP50(B)



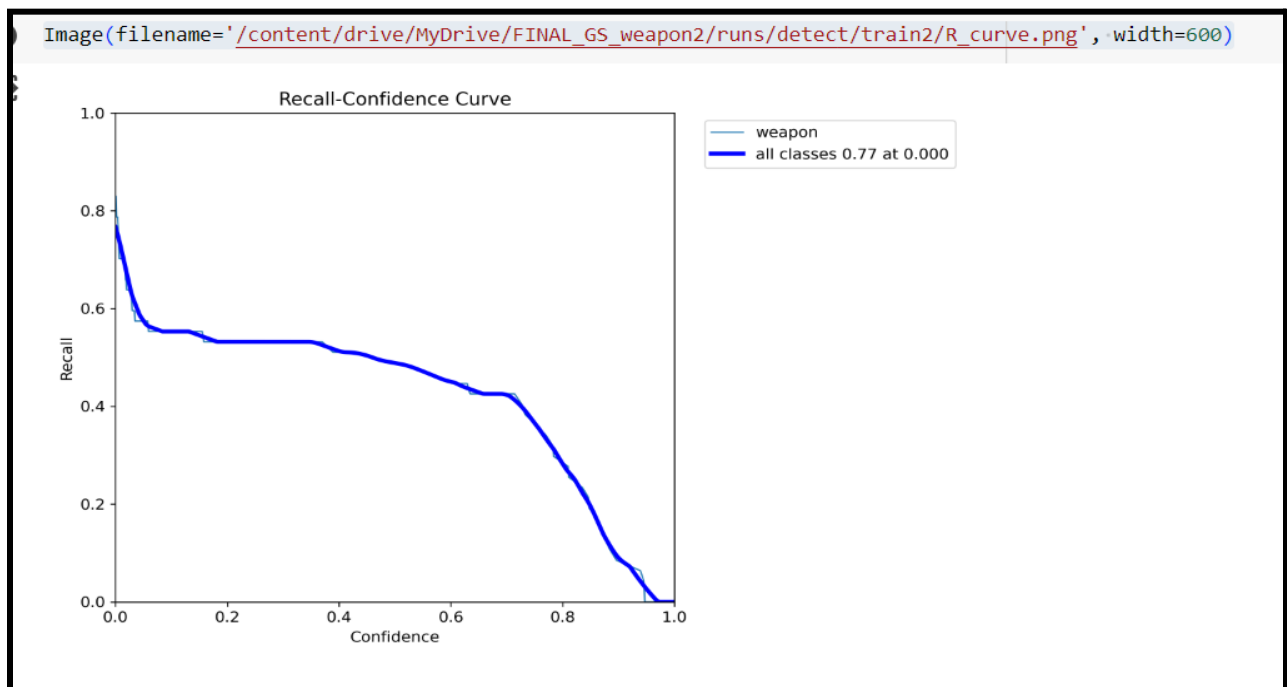
metrics/mAP50-95(B)



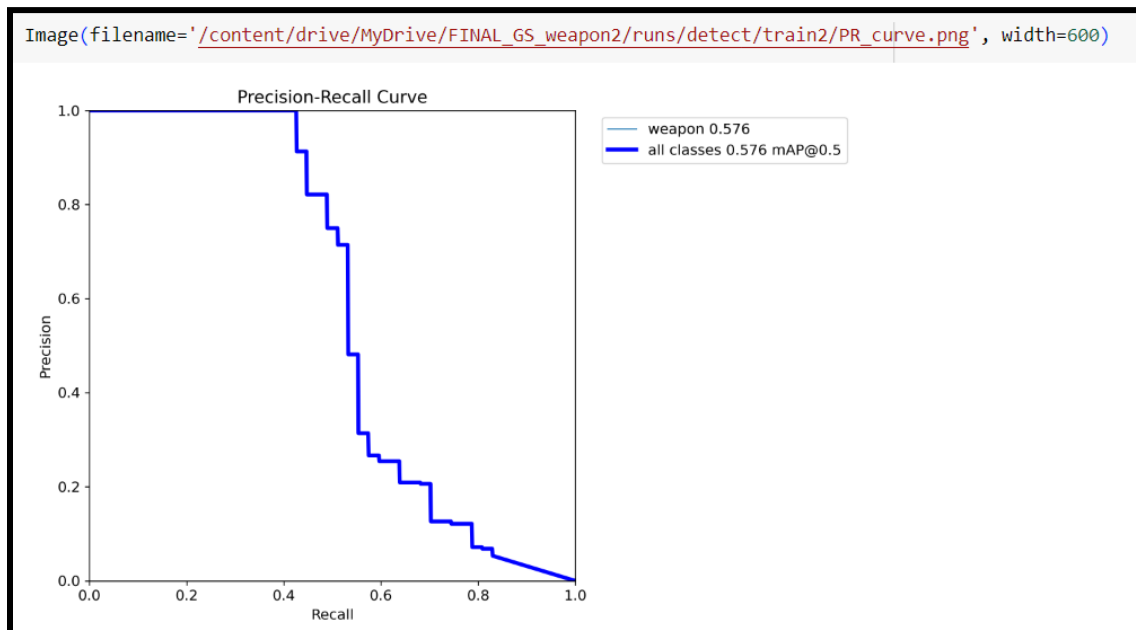
→ Precision-Confidence Curve



→ Recall-Confidence Curve



→ Precision-Recall Curve



→ Programmatic Predictions

In the final step, we load the YOLO model using the YOLO class and make predictions on a specific image programmatically.

```
0s ✓ model = YOLO('/content/drive/MyDrive/FINAL_GS_weapon2/runs/detect/train2/weights/best.pt')
```

[] Start coding or [generate](#) with AI.

